

Lecture 13. Factor Analysis

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Overview

- 1 Linear Algebra Review
- 2 Principal Components
- 3 Factor Analysis

Linear Algebra

Eigenvalues

Let $\mathbf{A}$ be a $p \times p$ matrix and $\mathbf{I}$ be the $p \times p$ identity matrix. Then the scalar quantities $\lambda_1, \dots, \lambda_p$ satisfying the polynomial equation $|\mathbf{A} - \lambda \mathbf{I}| = 0$ are called the eigenvalues of the matrix $\mathbf{A}$.

Linear Algebra

Eigenvectors

Let $\mathbf{A}$ be a $p \times p$ matrix and let λ be an eigenvalue of $\mathbf{A}$. If $\mathbf{x}$ is a nonzero $p \times 1$ vector such that $\mathbf{Ax} = \lambda\mathbf{x}$, then $\mathbf{x}$ is said to be an eigenvector of the matrix $\mathbf{A}$ associated with the eigenvalue λ .

Linear Algebra

Let's look at a 2x2 example:

$$\mathbf{A} = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$$

We need to solve for λ . The 2x2 case is simpler:

$$|\mathbf{A} - \lambda \mathbf{I}| = 0$$

The left-hand side within the determinant symbol is:

$$\begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} - \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix} = \begin{bmatrix} 2 - \lambda & 1 \\ 1 & 2 - \lambda \end{bmatrix}$$

Linear Algebra

Then, obtaining the determinant:

$$\det \begin{bmatrix} 2 - \lambda & 1 \\ 1 & 2 - \lambda \end{bmatrix} = (2 - \lambda)^2 - 1$$

This results in the following polynomial equation:

$$3 - 4\lambda + \lambda^2 = 0$$

Which has two solutions:

$$\lambda = 1 \text{ and } \lambda = 3$$

Linear Algebra

$$\mathbf{A} - \mathbf{1I} = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} - \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$$

Linear Algebra

For $\lambda = 1$:

$$(\mathbf{A} - \mathbf{1I})\mathbf{v} = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \begin{Bmatrix} v_1 \\ v_2 \end{Bmatrix} = \begin{Bmatrix} 0 \\ 0 \end{Bmatrix}$$

$$\mathbf{v} = \begin{Bmatrix} 1 \\ -1 \end{Bmatrix}$$

Linear Algebra

For $\lambda = 3$:

$$(\mathbf{A} - 3\mathbf{I})\mathbf{w} = \begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix} \begin{Bmatrix} w_1 \\ w_2 \end{Bmatrix} = \begin{Bmatrix} 0 \\ 0 \end{Bmatrix}$$

$$\mathbf{w} = \begin{Bmatrix} 1 \\ 1 \end{Bmatrix}$$

The vectors $\mathbf{v}$ and $\mathbf{w}$ are eigenvectors of $\mathbf{A}$ associated with the eigenvalues $\lambda = 1$ and $\lambda = 3$.

Linear Algebra

Spectral Decomposition Theorem

Let $\mathbf{A}$ be a $p \times p$ matrix, then $\mathbf{A}$ has p eigenvalues (λ_i) and associated eigenvectors ($\mathbf{e}_i$) which form pairs $(\lambda_1, \mathbf{e}_1), \dots, (\lambda_p, \mathbf{e}_p)$ where $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_p$, and the eigenvectors satisfy $\mathbf{e}_1' \mathbf{e}_1 = \mathbf{e}_2' \mathbf{e}_2 = \dots \mathbf{e}_p' \mathbf{e}_p = \mathbf{1}$ and $\mathbf{e}_i' \mathbf{e}_j = 0$ for $i \neq j$. We can write $\mathbf{A}$ in a spectral decomposition as:

$$\mathbf{A} = \sum_{i=1}^p \lambda_i \mathbf{e}_i \mathbf{e}_i'$$

Linear Algebra

A couple of other useful things...

The **trace** of a matrix is the sum of its diagonal elements,

$$tr(\mathbf{A}) = \sum_{i=1}^p Q_{ii}.$$

The trace is equal to the sum of the eigenvalues, $tr(\mathbf{A}) = \sum_{i=1}^p \lambda_i$.

A square matrix is said to be **orthogonal** if $\mathbf{Q}\mathbf{Q}' = \mathbf{Q}'\mathbf{Q} = \mathbf{I}$, or $\mathbf{Q}' = \mathbf{Q}^{-1}$.

Principal Components

Principal components is a **data reduction** technique.

The goal is to contain as much of the original information as possible in a smaller number of dimensions.

It may also help us understand the relationships in the data better.

Principal Components

Begin with a random vector $\mathbf{X}' = [X_1, X_2, \dots, X_p]$ that has covariance matrix Σ .

Define the following linear combinations:

$$Y_1 = \mathbf{a}'_1 \mathbf{X} = a_{11}X_1 + a_{12}X_2 + \dots + a_{1p}X_p$$

$$Y_2 = \mathbf{a}'_2 \mathbf{X} = a_{21}X_1 + a_{22}X_2 + \dots + a_{2p}X_p$$

$$\vdots$$

$$Y_p = \mathbf{a}'_p \mathbf{X} = a_{p1}X_1 + a_{p2}X_2 + \dots + a_{pp}X_p$$

Those $\mathbf{a}'_i$ for $i = 1$ to p are constant terms.

Principal Components

$$\begin{aligned} \text{Var}(Y_i) &= \mathbf{a}_i' \Sigma \mathbf{a}_i \\ \text{Cov}(Y_i, Y_k) &= \mathbf{a}_i' \Sigma \mathbf{a}_k \end{aligned}$$

The principal components are the uncorrelated linear combinations $Y_1, Y_2, \dots, Y_p$.

The first principal component is the linear combination $Y_1 = \mathbf{a}_1' \mathbf{X}$ that maximizes $\text{Var}(\mathbf{a}_1' \mathbf{X})$. Now, since we could always simply multiple $\mathbf{a}_1$ by a larger number, we also add the restriction that $\mathbf{a}_1' \mathbf{a}_1 = 1$.

The second principal component is the linear combination $Y_2 = \mathbf{a}_2' \mathbf{X}$ that maximizes $\text{Var}(\mathbf{a}_2' \mathbf{X})$ subject to the restriction that $\mathbf{a}_2' \mathbf{a}_2 = 1$ and $\text{Cov}(\mathbf{a}_1' \mathbf{X}, \mathbf{a}_2' \mathbf{X}) = 0$.

Principal Components

Let Σ be the covariance matrix associated with $\mathbf{X}$.

Let Σ have the eigenvalue-eigenvector pairs $(\lambda_1, \mathbf{e}_1), (\lambda_2, \mathbf{e}_2), \dots, (\lambda_p, \mathbf{e}_p)$.

Then the i^{th} principal component is $Y_i = \mathbf{e}_i' \mathbf{X}$.

$$\text{Var}(Y_i) = \mathbf{e}_i' \Sigma \mathbf{e}_i = \lambda_i \text{ and } \text{Cov}(Y_i, Y_k) = \mathbf{e}_i' \Sigma \mathbf{e}_k = 0.$$

Which means that the principal components are uncorrelated and have variances equal to the eigenvalues of Σ .

Principal Components

$$\begin{aligned}\sigma_{11} + \sigma_{22} + \cdots + \sigma_{pp} &= \sum_{i=1}^p \text{Var}(X_i) \\ &= \lambda_1 + \lambda_2 + \cdots + \lambda_p \\ &= \sum_{i=1}^p \text{Var}(Y_i)\end{aligned}$$

Since this is true, the proportion of variance accounted for by the i^{th} principal component Y_i is:

$$\frac{\lambda_i}{\lambda_1 + \lambda_2 + \cdots + \lambda_p}$$

Principal Components

It's also true that the components of the $\mathbf{e}_i$ vector has useful interpretations.

The magnitude of each e_{ik} measures the importance of the k^{th} variable to the i^{th} principal component.

Principal Components

Let's look at an example (from Johnson and Wichern). Start from the following covariance matrix for X_1 , X_2 , and X_3 :

$$\Sigma = \begin{bmatrix} 1 & -2 & 0 \\ -2 & 5 & 0 \\ 0 & 0 & 2 \end{bmatrix}$$

Principal Components

The eigenvalues and eigenvectors are:

$$\begin{aligned}\lambda_1 &= 5.83 & \mathbf{e}'_1 &= [0.383, -0.924, 0] \\ \lambda_2 &= 2.00 & \mathbf{e}'_2 &= [0, 0, 1] \\ \lambda_3 &= 0.17 & \mathbf{e}'_3 &= [0.924, 0.383, 0]\end{aligned}$$

Principal Components

The principal components are:

$$Y_1 = \mathbf{e}_1' \mathbf{X} = 0.383X_1 - 0.924X_2 + 0X_3$$

$$Y_2 = \mathbf{e}_2' \mathbf{X} = 0X_1 - 0X_2 + 1X_3$$

$$Y_3 = \mathbf{e}_3' \mathbf{X} = 0.924X_1 - 0.383X_2 + 0X_3$$

Principal Components

$$\sigma_{11} + \sigma_{22} + \sigma_{33} = 1 + 5 + 2 = 8$$

$$\lambda_1 + \lambda_2 + \lambda_3 = 5.83 + 2.00 + 0.17 = 8$$

In this example, the first principal component accounts for $5.83/8=72.9\%$ of the variance.

The first two components account for $7.83/8=97.9\%$ of the variance.

Principal Components

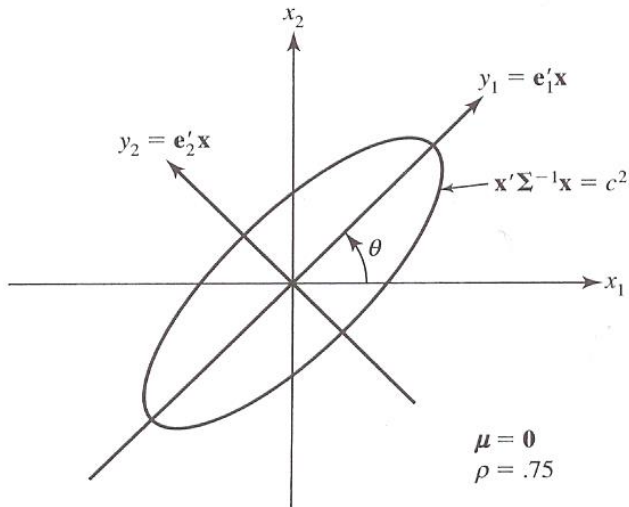
That's the algebraic approach to principal components.

There is also a geometric view.

Let's look at a bivariate example.

Principal Components

From Johnson and Wichern.



Principal Components

One caution, the principal components are based upon the variances. If the variances are on different scales, this can result in some variables predominating.

It is important to **standardize** the variables before finding principal components.

Principal Components

This is a useful data reduction technique.

Produces orthogonal variables that explain a large proportion of original variance.

Important to scale the variables before creating components.
Otherwise variables with “large” absolute variance will predominate.

Examples

R examples:

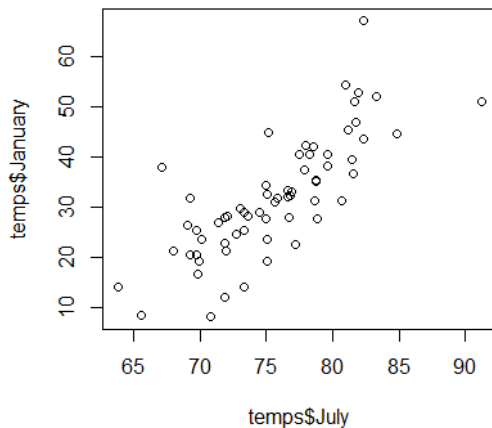
- First example looks at temperature in a variety of cities in January and July.
- Second example shows what happens when variables are not standardized.

Example 1

Temperature in January and July in 64 cities. Run the following R code to obtain the principal components:

```
library(car)
fit <- princomp(temps)
summary(fit) # print variance accounted for
loadings(fit) # pc loadings
plot(fit,type="lines") # scree plot
fit$scores # the principal components
```

Example 1



Example 1

Extract the eigenvalues:

```
> eig.val <- get_eigenvalue(fit)
> eig.val
eigenvalue variance.percent
Dim.1  1.7519238           88.98661
Dim.2  0.2168262          11.01339
cumulative.variance.percent
Dim.1           88.98661
Dim.2          100.00000
```

Example 1

Extract the eigenvalues:

```
> loadings(fit) # pc loadings
```

Loadings:

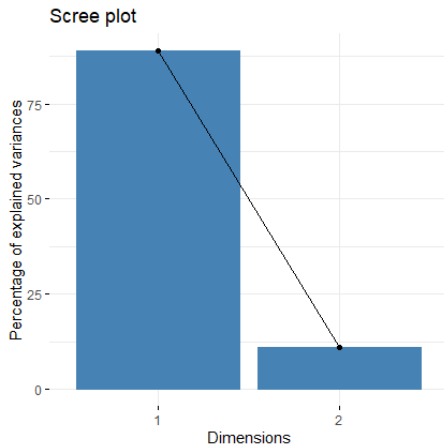
	Comp.1	Comp.2
January	0.707	0.707
July	0.707	-0.707

	Comp.1	Comp.2
SS loadings	1.0	1.0
Proportion Var	0.5	0.5
Cumulative Var	0.5	1.0

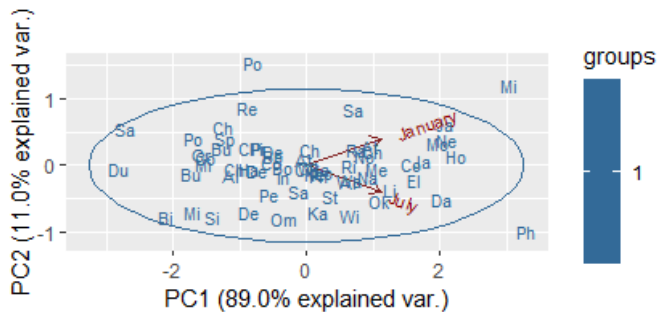
$$Y_1 = 0.707 \times \text{January} + 0.707 \times \text{July}$$

$$Y_2 = 0.707 \times \text{January} - 0.707 \times \text{July}$$

Example 1



Example 1



Example 1

R code:

```
library(car)
fit <- princomp(temps)
summary(fit) # print variance accounted for
loadings(fit) # pc loadings
plot(fit,type="lines") # scree plot
fit$scores # the principal components
plot(fit$scores)
```

Example 2

Taken from Everitt and Hothorn. They report on a study that took eight blood chemistry variables. The measurements are from 72 patients in a clinical trial.

Here are the standard deviations:

```
> blood_sd
rblood  plate wblood   neut   lymph   bilir  sodium  potass
 0.371  41.253   1.935   0.077   0.071   4.037   2.732   0.297
```

Note that `plate` has a very large SD relative to other variables.

Example 2

If I use the covariance matrix as the input for selecting principal components, I get:

```
> blood_pcacov <- princomp(covmat = blood_cov)
> summary(blood_pcacov, loadings = TRUE)
Importance of components:
```

	Comp.1	Comp.2	Comp.3	Comp.4	Comp.5	Comp.6	Comp.7	Comp.8
Standard deviation	41.29	3.8802	2.642	1.6246	3.5e-01	2.6e-01	8.5e-02	2.4e-02
Proportion of Variance	0.99	0.0087	0.004	0.0015	7.2e-05	3.8e-05	4.2e-06	3.3e-07
Cumulative Proportion	0.99	0.9943	0.998	0.9999	1.0e+00	1.0e+00	1.0e+00	1.0e+00

Loadings:

	Comp.1	Comp.2	Comp.3	Comp.4	Comp.5	Comp.6	Comp.7	Comp.8
rblood					0.943	0.329		
plate	-0.999							
wblood	-0.192		-0.981					
neut						0.758	0.650	
lymph						-0.649	0.760	
bilir	0.961	0.195	-0.191					
sodium	0.193	-0.979						
potass				0.329	-0.942			

Example 2

If I use the correlation matrix as the input for selecting principal components, I get:

```
> blood_pcacor <- princomp(covmat = blood_corr)
> summary(blood_pcacor, loadings = TRUE)
```

Importance of components:

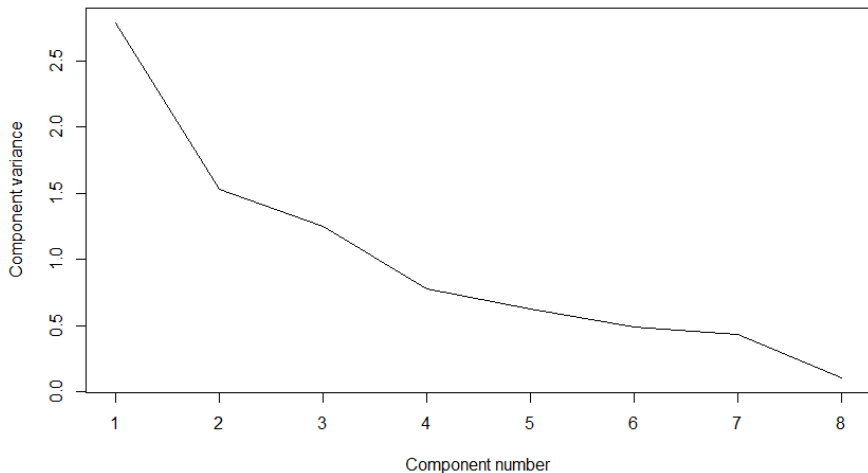
	Comp.1	Comp.2	Comp.3	Comp.4	Comp.5	Comp.6	Comp.7	Comp.8
Standard deviation	1.67	1.24	1.12	0.882	0.788	0.699	0.660	0.320
Proportion of Variance	0.35	0.19	0.16	0.097	0.078	0.061	0.054	0.013
Cumulative Proportion	0.35	0.54	0.70	0.794	0.872	0.933	0.987	1.000

Loadings:

	Comp.1	Comp.2	Comp.3	Comp.4	Comp.5	Comp.6	Comp.7	Comp.8
[1,]	-0.194	0.417	0.400	0.652	0.175	-0.363	0.176	0.102
[2,]	-0.400	0.154	0.168		-0.848	0.230	-0.110	
[3,]	-0.459		0.168	-0.274	0.251	0.403	0.677	
[4,]	-0.430	-0.472	-0.171	0.169	0.118		-0.237	0.678
[5,]	0.494	0.360		-0.180	-0.139	0.136	0.157	0.724
[6,]	0.319	-0.320	-0.277	0.633	-0.162	0.384	0.377	
[7,]	0.177	-0.535	0.410	-0.163	-0.299	-0.513	0.367	
[8,]	0.171	-0.245	0.709		0.198	0.469	-0.376	

Example 2

Scree diagram



Uses of Principal Components

One use is to understand the data better.

In regression analysis when prediction is the goal...

- Creating predictors when p is large relative to n
- When predictors are highly correlated

Factor Analysis

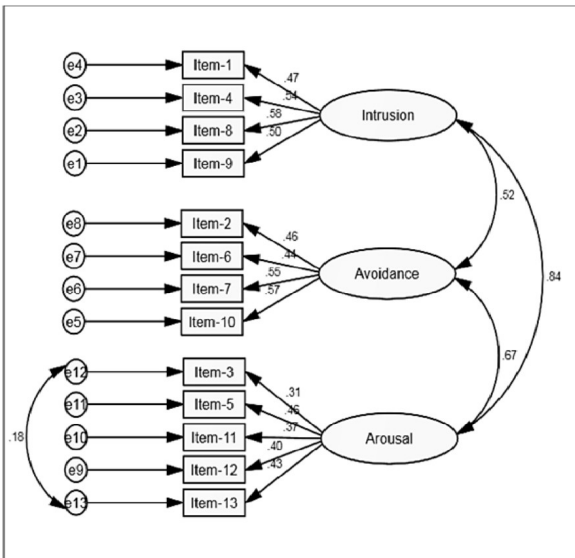
Factor analysis is a technique related to principal components.

The goal is to explain relationships among variables in terms of underlying variables known as **factors**.

Group things together that are highly correlated. Assume they are driven by the same underlying factor.

Very common approach in psychology. Take multiple measurements on an assumed underlying characteristic of personality.

Factor Analysis



Factor Analysis

Notation for the orthogonal factor model.

Observed random vector $\mathbf{X}$ with p components. That is, p variables in my data.

This vector comes from a $N_p(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ distribution.

We also have m factors, $F_1, F_2, \dots, F_m$ and p errors $\epsilon_1, \epsilon_2, \dots, \epsilon_p$.

Factor Analysis

With this notation we can write the following model:

$$X_1 - \mu_1 = \ell_{11}F_1 + \ell_{12}F_2 + \cdots + \ell_{1m}F_m + \epsilon_1$$

$$X_2 - \mu_2 = \ell_{21}F_1 + \ell_{22}F_2 + \cdots + \ell_{2m}F_m + \epsilon_2$$

$$\vdots$$

$$X_p - \mu_p = \ell_{p1}F_1 + \ell_{p2}F_2 + \cdots + \ell_{pm}F_m + \epsilon_p$$

Or, in matrix notation:

$$\mathbf{X} - \boldsymbol{\mu} = \mathbf{LF} + \boldsymbol{\epsilon}$$

Factor Analysis

We can rearrange the terms:

$$\mathbf{X} = \boldsymbol{\mu} + \mathbf{LF} + \boldsymbol{\epsilon}$$

This may look like a regression model, but the only observed quantities are $\mathbf{X}$ and $\boldsymbol{\mu}$.

Need further assumptions to motivate such a model.

Factor Analysis

Additional assumptions are:

$$E(\mathbf{F}) = \mathbf{0}$$

$$\text{Cov}(\mathbf{F}) = \mathbf{I}$$

$$E(\epsilon) = \mathbf{0}$$

$$\text{Cov}(\epsilon) = \Psi$$

$$\text{Cov}(\epsilon, \mathbf{F}) = \mathbf{0}$$

where Ψ is a diagonal matrix (ϵ_i terms are not correlated).

Factor Analysis

This model then implies a covariance structure for $\mathbf{X}$.

$$\text{Cov}(\mathbf{X}) = \mathbf{LL}' + \Psi$$

$$\text{Var}(X_i) = \ell_{i1}^2 + \cdots + \ell_{im}^2 + \psi_i$$

$$\text{Cov}(X_i, X_k) = \ell_{i1}\ell_{k1} + \cdots + \ell_{im}\ell_{km}$$

where $\psi_i = \text{Var}(\epsilon_i)$.

Factor Analysis

Also under this model, it is possible to partition the variance for each X_i .

The portion of the variance due to the factors is called the **communality**.

$$\sigma_{ii} = \ell_{i1}^2 + \ell_{i2}^2 + \cdots + \ell_{im}^2 + \psi_i$$

This is sometimes denoted h_i^2 :

$$h_i^2 = \ell_{i1}^2 + \ell_{i2}^2 + \cdots + \ell_{im}^2$$

Factor Analysis

Let's look at an example:

$$\Sigma = \begin{bmatrix} 19 & 30 & 2 & 12 \\ 30 & 57 & 5 & 23 \\ 2 & 5 & 38 & 47 \\ 12 & 23 & 47 & 68 \end{bmatrix}$$

Then we can use the formula $\Sigma = \mathbf{LL}' + \Psi$.

Factor Analysis

$$\begin{bmatrix} 19 & 30 & 2 & 12 \\ 30 & 57 & 5 & 23 \\ 2 & 5 & 38 & 47 \\ 12 & 23 & 47 & 68 \end{bmatrix} = \begin{bmatrix} 4 & 1 \\ 7 & 2 \\ -1 & 6 \\ 1 & 8 \end{bmatrix} \begin{bmatrix} 4 & 7 & -1 & 1 \\ 1 & 2 & 6 & 8 \end{bmatrix} + \begin{bmatrix} 2 & 0 & 0 & 0 \\ 0 & 4 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 3 \end{bmatrix}$$

In this case, $h_1^2 = 4^2 + 1^2 = 17$. And the total variance of X_1 is $17 + 2 = 19$.

Factor Analysis

There is a remaining issue to discuss. The solutions to any factor model aren't unique.

$$\mathbf{X} - \mu = \mathbf{L}\mathbf{F} + \epsilon = \mathbf{L}\mathbf{T}\mathbf{T}'\mathbf{F} + \epsilon = \mathbf{L}^*\mathbf{F}^* + \epsilon$$

The non-uniqueness is apparent in the above equality. This doesn't change the communalities.

The $\mathbf{L}$ matrix is multiplied by some $\mathbf{T}$ that aids in interpretation of the loadings. This is called factor "rotation."

Because of this relationship, the scale of the covariance matrix doesn't matter either. Factor analysis is scale invariant (principal components are not).

Factor Analysis

Two different *types* of rotation are possible.

- **Orthogonal.** Factors are restricted to be uncorrelated. E.g. Varimax.
- **Oblique.** Factors are allowed to be correlated. E.g. Oblimin

Factor Analysis

Orthogonal rotations are useful for producing factors that are generalizable. Easier to label and interpret.

Oblique rotations are useful for obtaining “best fit.”

Factor Analysis

The number of factors may be chosen in different ways. With ML estimation, a test of whether a specified number of factors that are adequate to account for the observed covariances is available.

The `psych` package in R will use this method to identify the number of factors.

Factor Analysis

```
library(psych)
fit2 <- principal(beer, nfactors=2, rotate="varimax")
summary(fit2) # print variance accounted for
loadings(fit2) # loadings
```

Factor Analysis

Here are the factors:

```
> loadings(fit3)
```

Loadings:

	RC1	RC2
cost	0.109	0.954
size	-0.201	0.956
alcohol	0.218	0.939
reputat	-0.723	-0.204
color	0.900	0.216
aroma	0.930	-0.130
taste	0.950	-0.118

Factor Analysis vs Principal Components

Factor Analysis	Principal Components
Explain <i>covariances</i> of observed variables.	Explain <i>variance</i> of observed variables.
Adding factors will change the loadings of all factors.	Adding components will not change previous components.
Factor scores requires several decisions by the analyst (rotation, number of factors).	Principal components are straightforward to calculate. Choose number via scree plot.
Factor analysis scale invariant.	Principal components may change depending upon use of covariance or correlation matrix.